

Rezultate și indicații la exercițiile propuse

Capitolul 1

$$1.1 \quad n \cong N_D = 9 \cdot 10^{15} \text{ cm}^{-3}; \quad p = 0.25 \cdot 10^5 \text{ cm}^{-3}$$

$$1.2 \quad p = 4 \cdot 10^{19} \text{ cm}^{-3}$$

$$1.3 \quad \text{a) } p \cong N_A = 3 \cdot 10^{15} \text{ cm}^{-3}; \quad n = 7.5 \cdot 10^4 \text{ cm}^{-3}; \quad \rho = 4.6 \, \Omega \cdot \text{cm}$$

$$\text{b) } p = 3 \cdot 10^{15} \text{ cm}^{-3}; \quad n = 7.5 \cdot 10^4 \text{ cm}^{-3}; \quad \rho = 6.9 \, \Omega \cdot \text{cm}$$

$$1.5 \quad R_T = 1280 \, \Omega; \quad NTC = -53.3 \frac{\Omega}{K}$$

$$1.6 \quad J_{drift} = 152 \frac{A}{cm^2}$$

$$1.7 \quad v_{drift} = 4000 \text{ cm/s}; \quad D = 10 \frac{cm^2}{s}$$

1.8

$$\rho = \frac{1}{q \cdot (n \mu_n + p \mu_p)}; \quad n = n_0 + N \cdot \tau; \quad p \cong p_0; \quad n_0 = \frac{1}{\rho_0 q n};$$

$$\rho \cong 0.35 \, \Omega \cdot \text{cm}; \quad \frac{\sigma}{\sigma_0} \cong 28.6$$

$$1.9 \quad \text{a) } 16.67 \text{ lm/W}; \text{ b) } 79.6 \text{ cd}; \text{ c) } 35.36 \text{ lx}; \text{ d) } E_v = \frac{I_v}{r^2}.$$